

Gravitational Wave Astrophysics using Rubin Observatory after the LSST

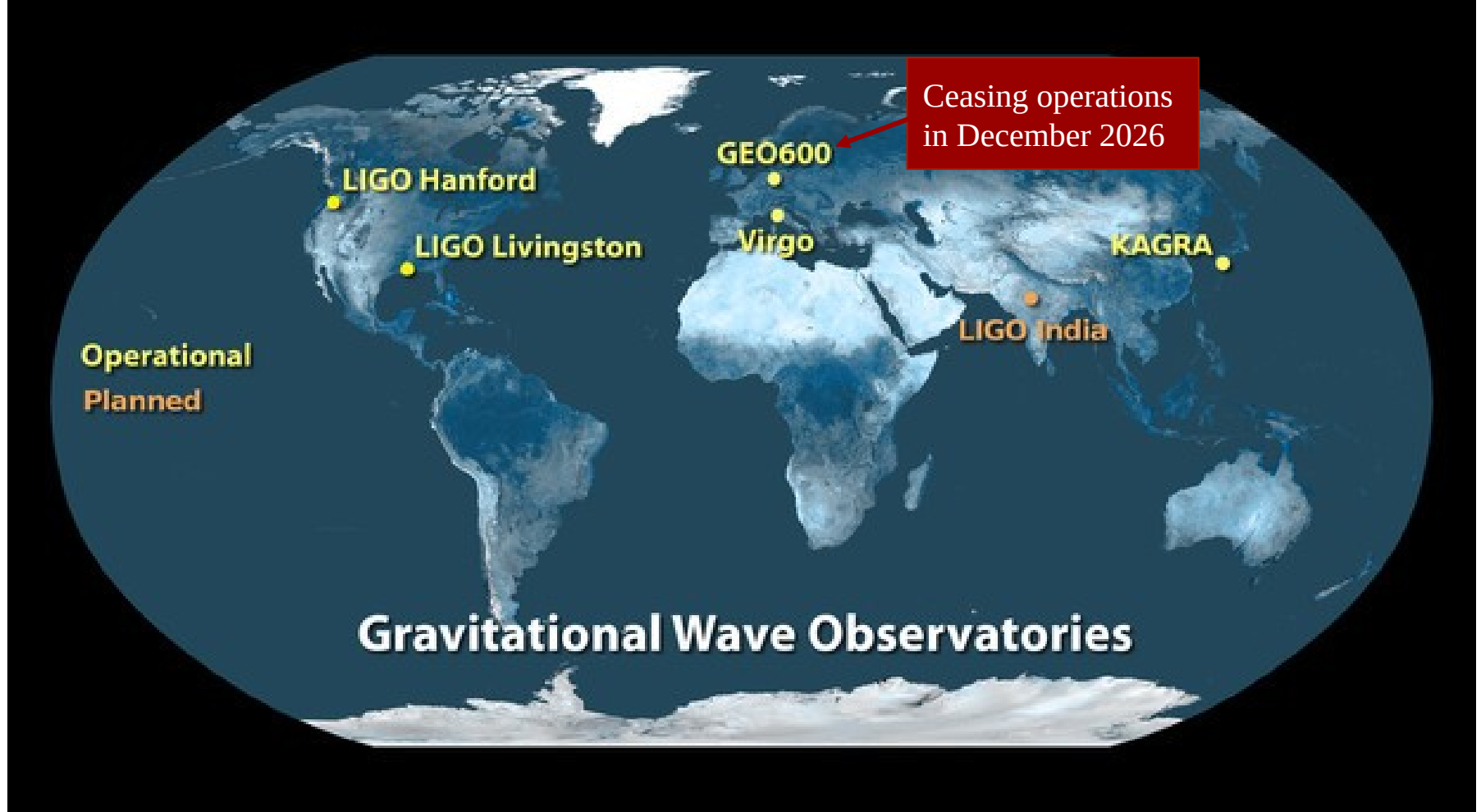
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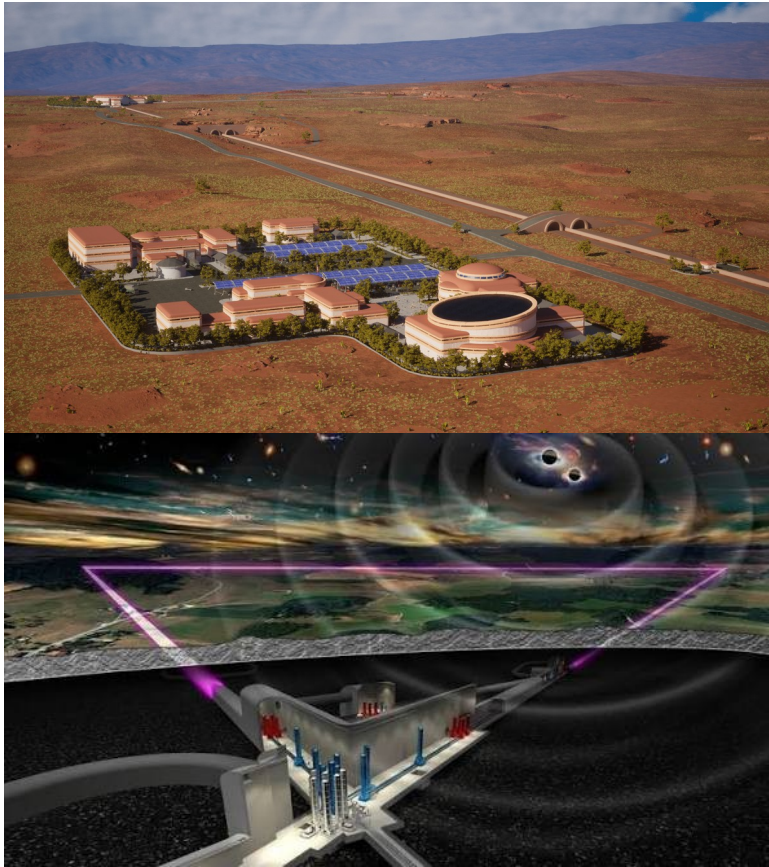
Rubin 2036 Themes

Participants join one of five thematic groups (A-E):

- A) Continued LSST-style surveying.
- B) Alternative cadences & footprints.
- C) Narrow- & medium-band surveys, other filters, filter offsets.
- D) Targets of opportunity & multi-messenger astronomy.
- E) Enabling technology & upgrades.

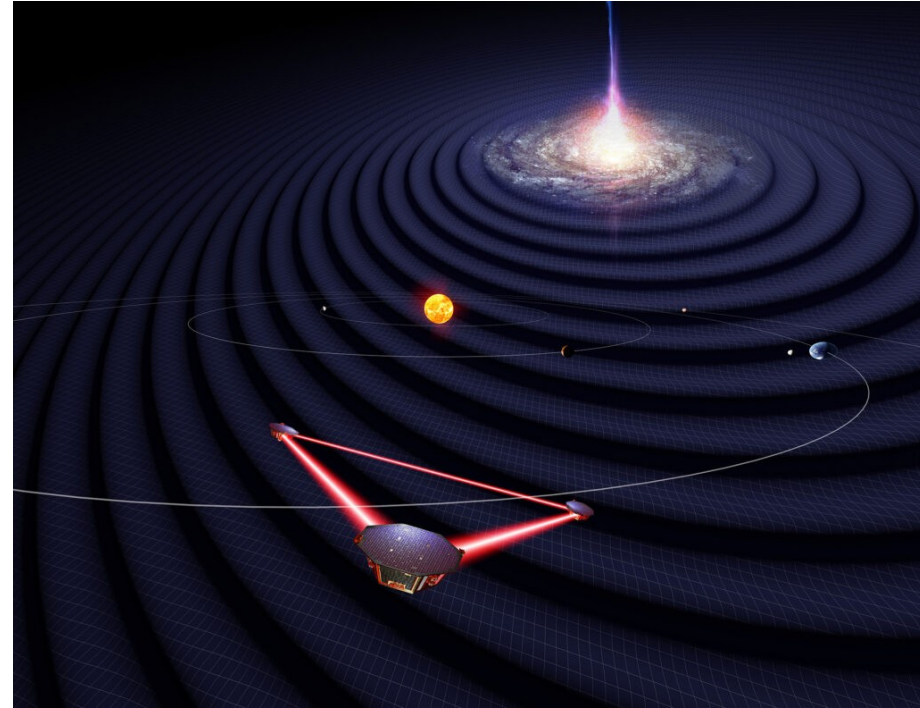


Cosmic Explorer - Ready in 2040



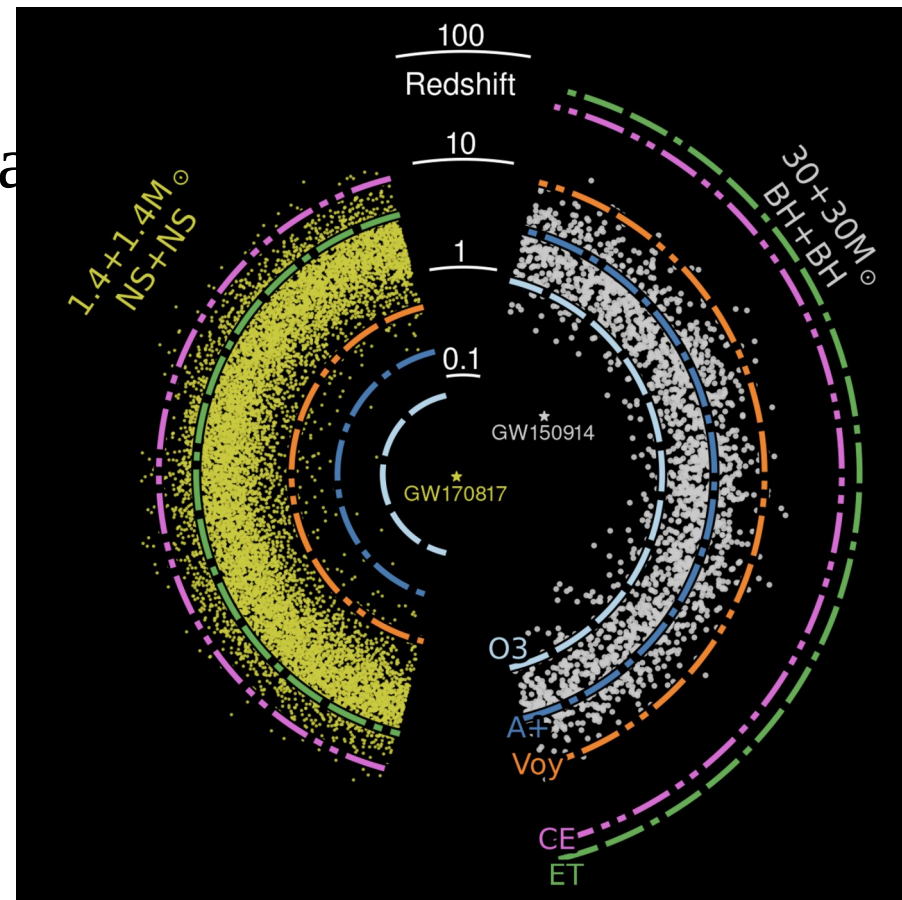
Einstein Telescope - Ready in 2035

Laser Interferometer Space Antenna - Launch in 2035

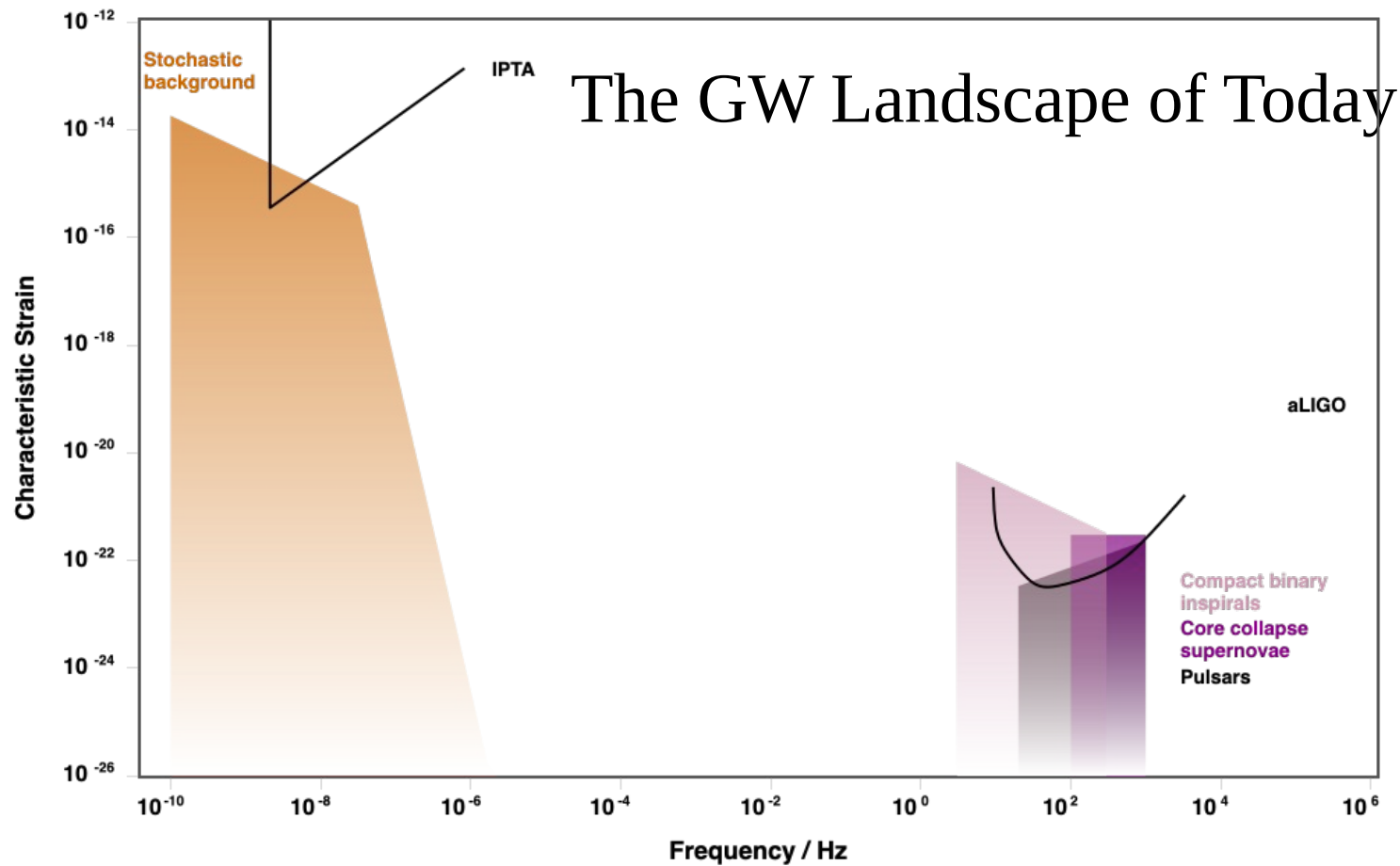


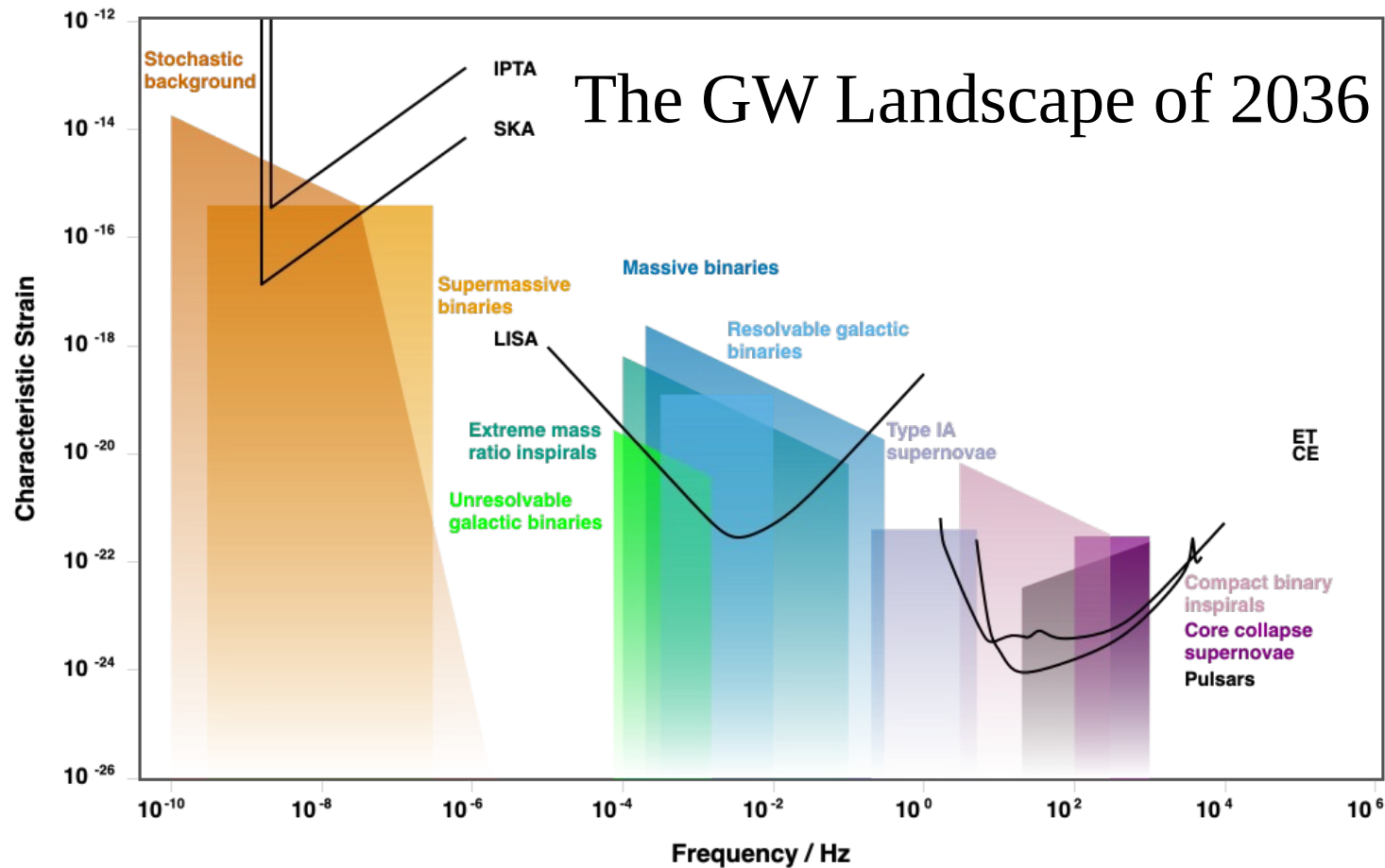
New Sensitivity at Cosmic Scale

- Current GW detectors probe a small fraction of the observable universe.
- Future observations will detect binary neutron star mergers out to $z \sim 10$.
- Electromagnetic telescopes are needed to capitalize on a wealth of new GW data.



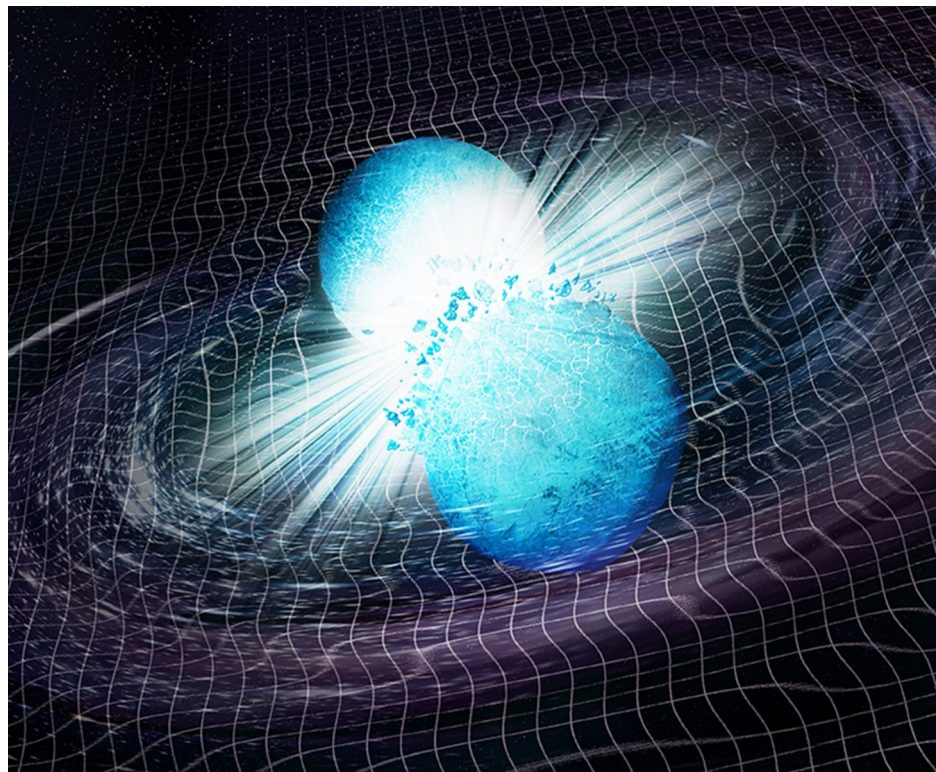
Astrophysical horizon of current and proposed future detectors for compact binary systems [2].





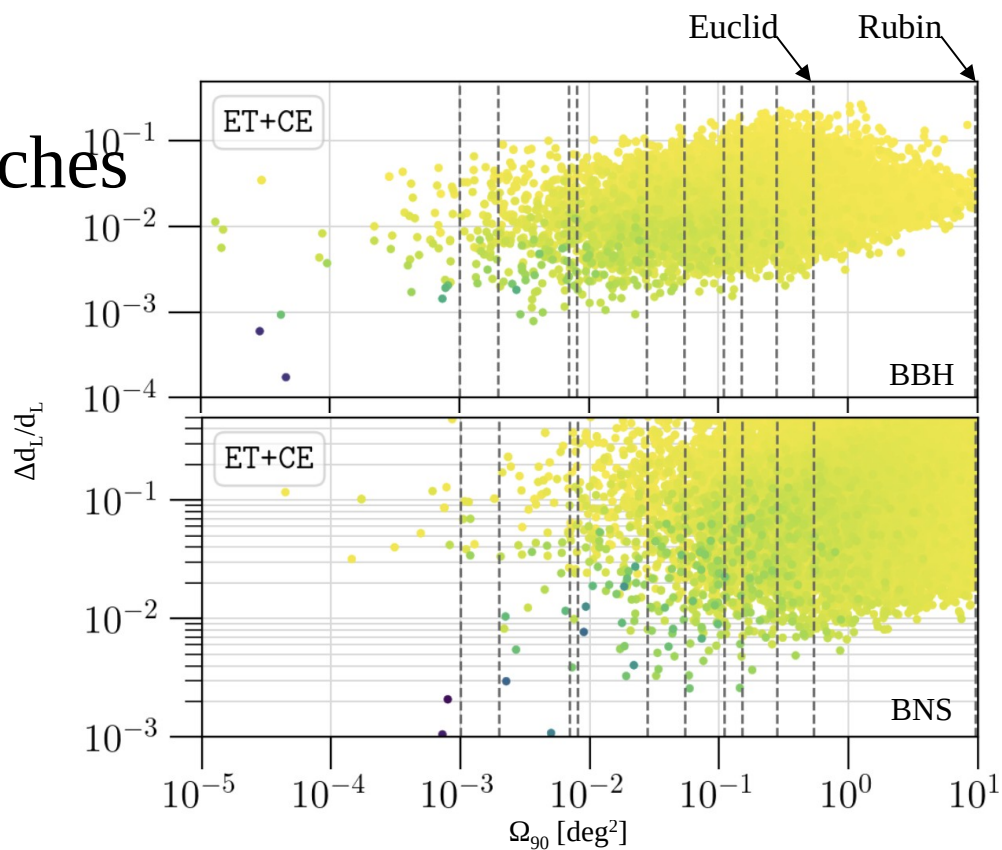
2036 Science in GW Astrophysics with Rubin Observatory

- **Bright siren detection:** Mergers at higher redshift from:
 - Binary neutron stars detected by Cosmic Explorer and/or Einstein Telescope.
 - Massive black hole binaries detected by LISA.
- **Pre-merger monitoring:** enabled by LISA detection of massive black hole binaries.
- **Host galaxy environments:** Correlating GW source properties with their host galaxy properties to reduce systematics for dark siren measurements.
- **Modified GW propagation:** High redshift sources identified from Cosmic Explorer and/or Einstein Telescope signals.



Rubin is Ideal for GW Searches

- Rubin FOV is highly compatible with well-localized BNS and BBH mergers from Cosmic Explorer and/or Einstein Telescope.
- Inefficient use for the best localized events, but can complement other narrow FOV facilities.



The 90% localization area and fractional luminosity distance of forecasted Cosmic Explorer BBH and BNS mergers. From [2].

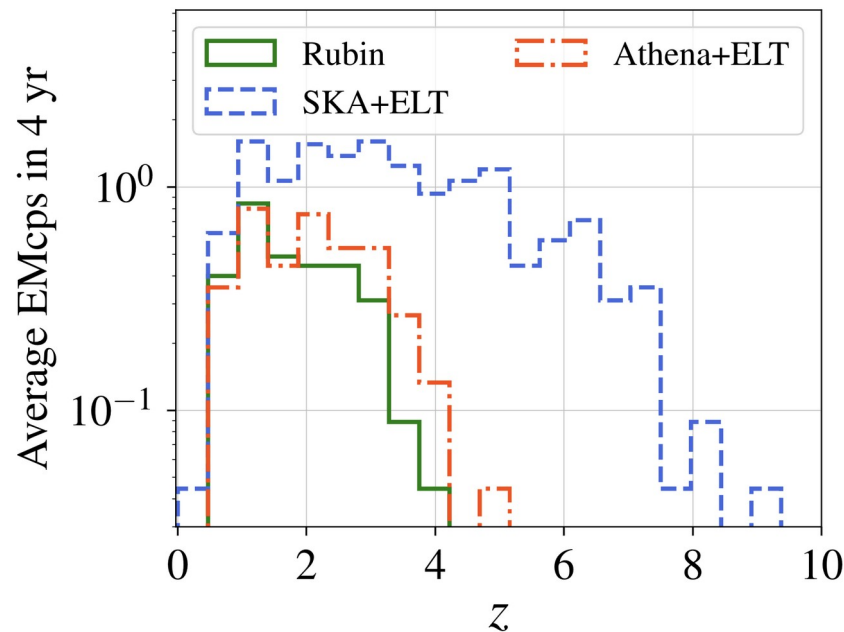
Can Rubin Detect Any of the Electromagnetic Counterparts?

Massive Black Hole Binaries from LISA

- Detectable by Rubin out to $z \sim 4$.
- Horizon of detection is highly dependent on AGN disk model.

Binary neutron stars detected by CE + ET

- Rubin can detect the electromagnetic counterparts to ~ 100 binary neutron stars per year with Einstein Telescope alone.
- Cosmic explorer will enable more detections and better localization.



Average number of electromagnetic counterparts detected in 4 years by different instruments, associated with LISA massive black hole binary detection, using the Q3d emission model [3,4].

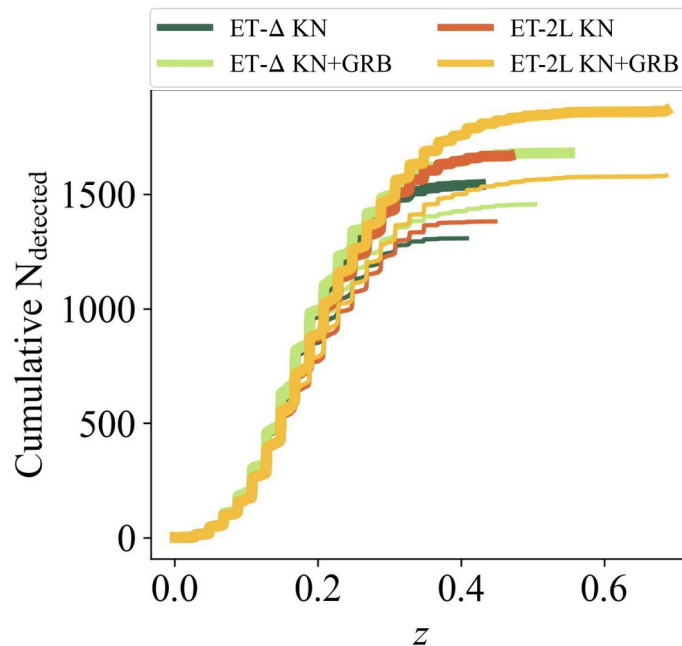
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Massive Black Hole Binaries from LISA

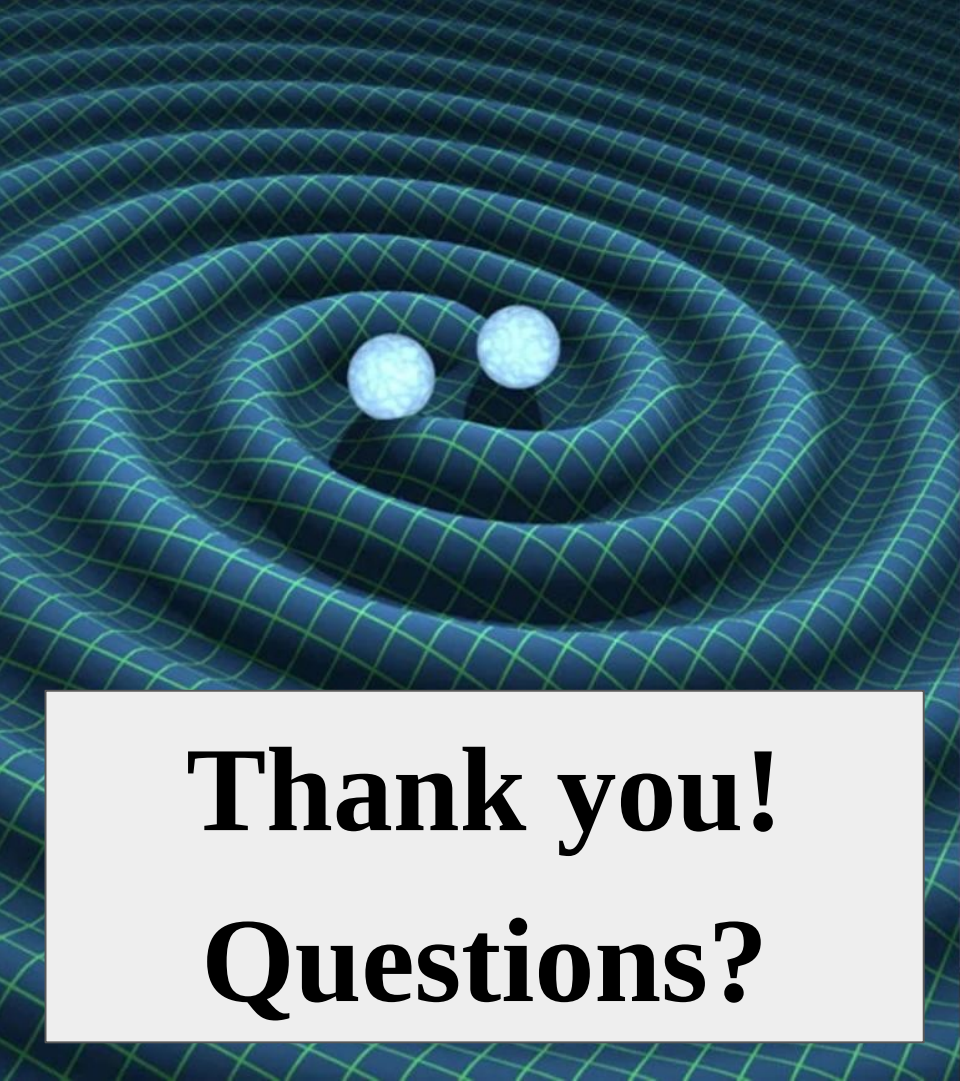
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Cumulative number of KN and KN+GRB counterpart detections by Rubin Observatory from Einstein Telescope observations over ten years of operations. Thick lines indicate the 1200s exposure time, and the thin lines indicate 600s exposures [5].



Gravitational waves provide a calibration free probe of cosmological parameters and extreme astrophysics.

Rubin Observatory's discovery capabilities make it well-suited to detect electromagnetic counterparts to a wealth of GW data in the post-LSST era.

This discovery capability will enable a wide range of cosmological and astrophysical studies of the universe.

Thank you!
Questions?



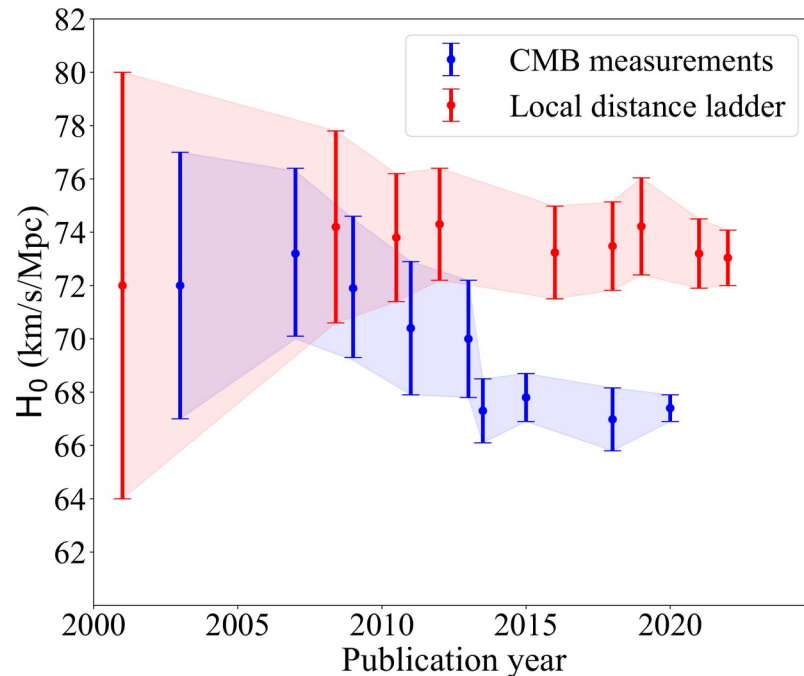
References

1. Determining the Hubble Constant from Gravitational Wave Observations, Bernard Schutz, Nature, Volume 323, Issue 6086
2. A Horizon Study for Cosmic Explorer: Science, Observatories, and Community, Evans et al, arXiv:2109.09882
3. Massive black hole binaries in LISA: Multimessenger prospects and electromagnetic counterparts, Mangiagli et al, Physical Review D, Volume 106, Issue 10, article id.103017
4. Formation of supermassive black holes by direct collapse in pre-galactic haloes, Begelman et al, Monthly Notices of the Royal Astronomical Society, Volume 370, Issue 1
5. Prospects for optical detections from binary neutron star mergers with the next-generation multi-messenger observatories, Loffredo et al, arXiv:2411.02342

Extra slides

The Λ CDM Cosmological Model

- The Λ CDM cosmological model is supported by a wealth of observational data.
- Two recent results indicate that Λ CDM may be incomplete:
 - DESI constraints on $w_0 w_a$ dark energy.
 - H_0 tension.
- A $>5\sigma$ discrepancy in measurements of H_0 has emerged, using two different probes:
 - Local distance ladder measurements [1].
 - Cosmic Microwave Background [2].
- This discrepancy is one of the most pressing questions for modern cosmology.



The precision of H_0 measurements over time, separated by measurement probe.

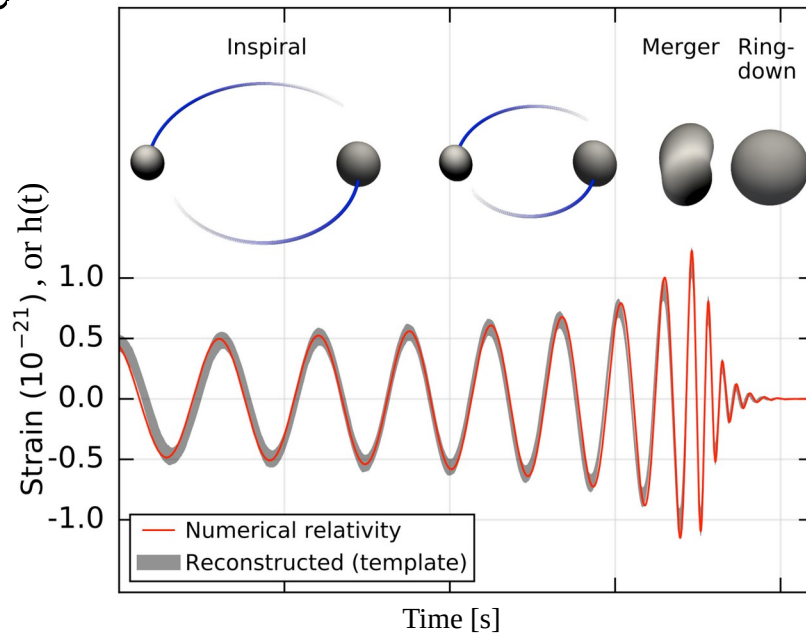
Gravitational Wave Cosmology With Standard Sirens

From GW mergers, we can directly measure luminosity distance [1]:

$$d_L \propto \frac{f(t)}{f(t)^3 h(t)}$$

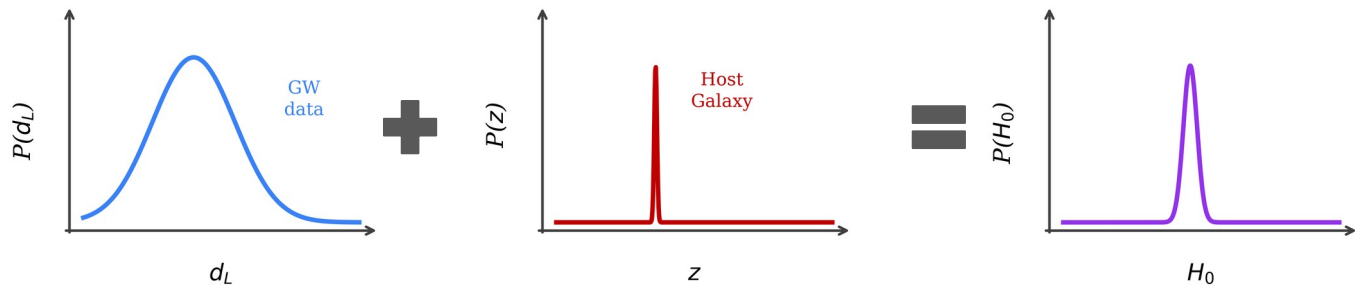
Luminosity distances from GW binary mergers help us understand cosmology and measure H_0 .

$$d_L = \frac{c(1+z)}{H_0} \int_0^z \frac{dz'}{\sqrt{(1+z')^3 \Omega_M + \Omega_\Lambda}}$$

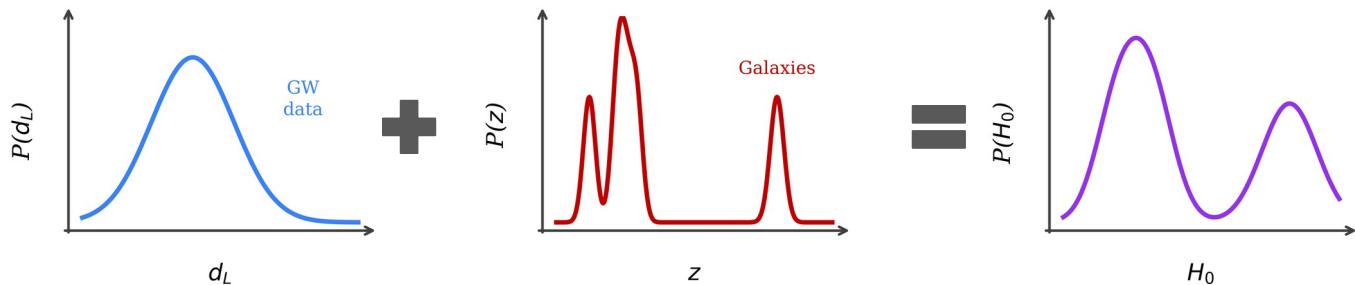


An illustration of the first ever gravitational wave detection, GW150914.

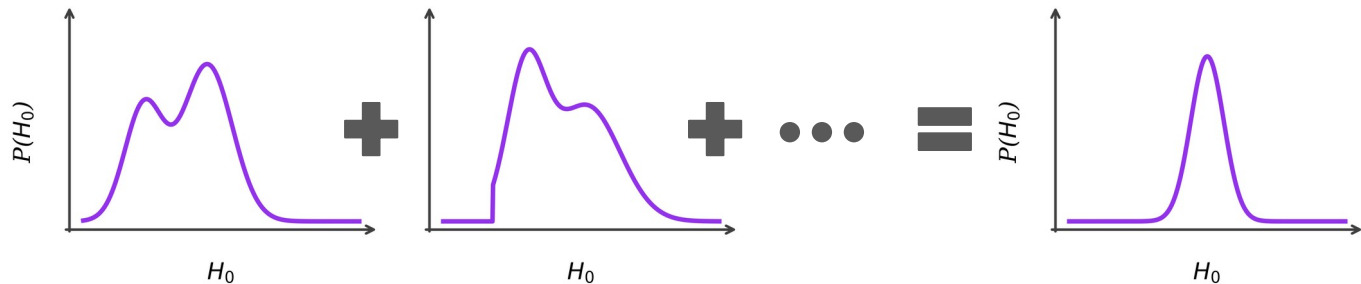
For one **bright siren**,
where a counterpart
to a GW is detected:



For one **dark siren**,
where no counterpart
to a GW is detected:



For many dark
sirens:



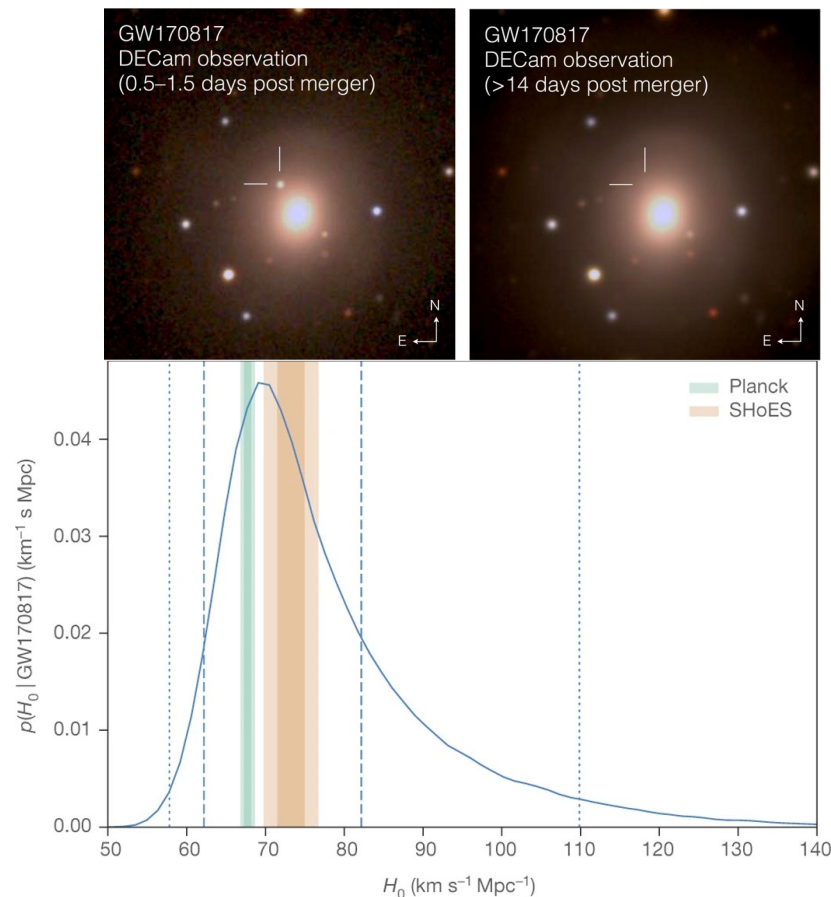
Bright Sirens - GW170817

- On August 17, 2017, binary neutron star merger was detected by LVK detectors [4,5].

- Within 24 hours, observations across the EM spectrum identified a kilonovae (KNe) in NGC 4993

$$H_0^{[6,7,8]} = 70_{-8}^{+12} \text{ km s}^{-1} \text{ Mpc}^{-1}$$

- The first bright siren measurement constrained H_0 to 14% precision.



Top: The kilonova AT2017gfo associated with GW170817, detected by the Dark Energy Camera (DECAM) [7]. Bottom: The resulting H_0 posterior of GW170817, when using the redshift of NGC 4993 [5].

Bright Sirens Since GW170817

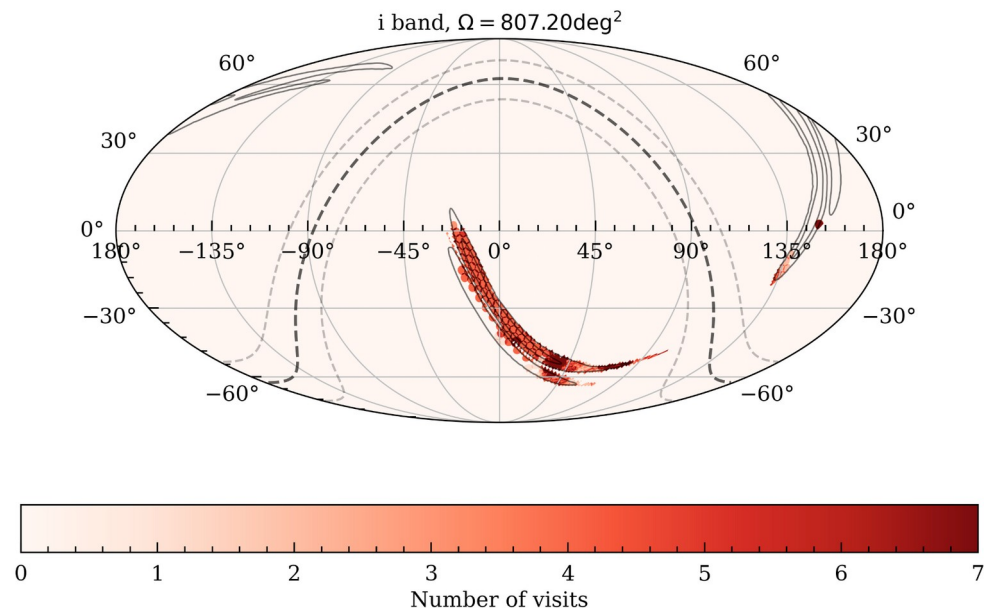
- Two long gravitational wave observing runs have concluded since 2017.
- Significant telescope time has been dedicated to identifying the next bright siren.
- Despite efforts by large consortiums (GROWTH, GW-MMADS, DESGW, etc.), no additional bright sirens have been identified.
- Why is there a lack of bright sirens?
 - Underestimated rates for GWs.
 - Optical sources are fainter than expected.
 - Emission timescales may differ from GW170817.



The Víctor M. Blanco 4-meter telescope (foreground), which hosts DECAM as its primary instrument, on the summit of Cerro Tololo. DECAM has served as an ideal search-and-discovery instrument for electromagnetic counterparts to gravitational waves, until the LSST began.

Bright Sirens With LSST

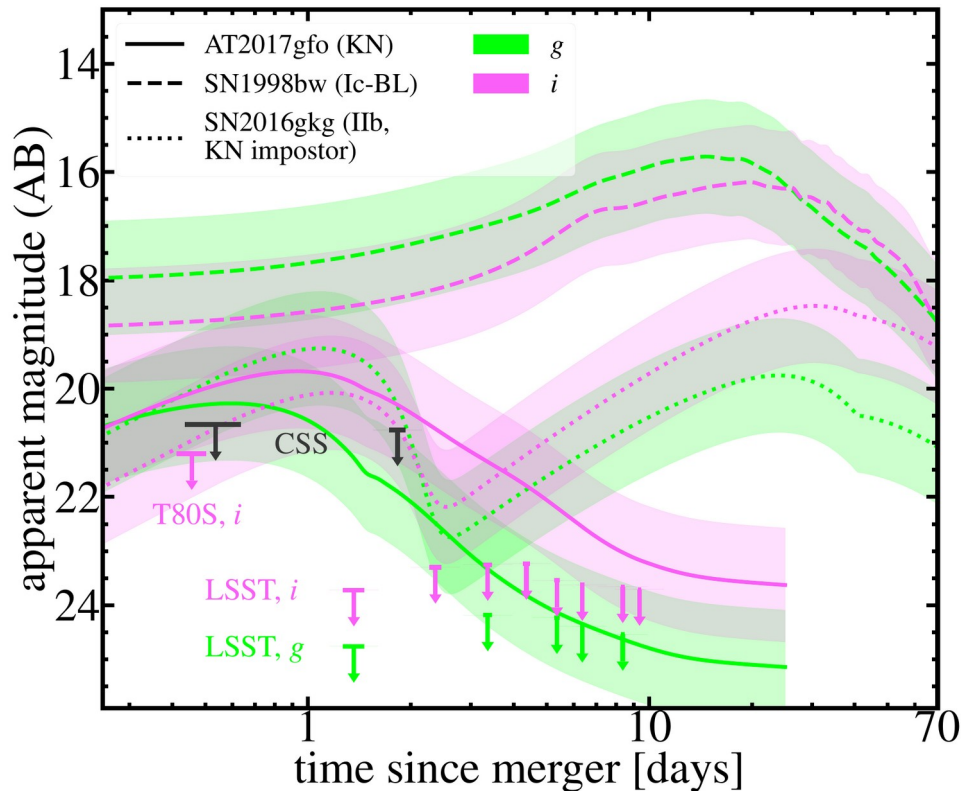
- GW follow up is a large part of the Rubin Target-of-Opportunity (ToO) program.
- The field-of-view of LSSTCam allows for rapid scanning of the sky compared to other telescopes.
- Rubin's depth allows for discovery of fainter sources.
- Several GW ToOs have already been activated during early Rubin operations.
- Preparations for the upcoming GW observing run are underway, which will start in October 2026.



The number of visits in i-band on the night of November 14th, 2025, from the ToO-triggered search for an electromagnetic counterpart to GW merger candidate S251112cm. The 50% and 90% contours of the GW merger candidate S251112cm area shown. All visits achieved within 3 hours, with similar area coverage in g band during that time period.

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Depths achieved in the search for an electromagnetic counterpart to S251112cm. Observations from LSST, the T80-South telescope, and Catalina Sky Survey (CSS) are shown. Models for a kilonova, a kilonova impostor, and a supernova Ic at the GW luminosity distance are also shown. LSST depths achieve detection at a timescale previously inaccessible for wide-area searches. From [9].